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Syrian Virtual University



Bachelor of IT Program BAIT

MS SQL Server Administration IIS404

Chapter six

Performance Monitoring and Tuning in MS SQL Server

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1. Intro

- لمراقبة العمليات التي تتم من قبل المستخدمين على ال DB نستطيع الإستعانة بتطبيقات خارجية مستقلة (third party applications) تكلف كثيرا و تقوم بمراقبة جميع العمليات ضمن ال DB أو إستخدام أدوات المراقبة الموجودة بشكل أساسي ضمن ال SQL Server و من أهمها:

Another mission of a DBA is monitoring the database, Microsoft SQL Server provides a comprehensive set of tools for monitoring events in SQL Server and for tuning the physical database design. The choice of tool depends on the type of monitoring or tuning to be done and the particular events to be monitored. There're a lot of techniques and third-party applications and they are in continual development, but here we will list some as an initial entry and then you can discover more.

2. SQL Server monitoring and tuning tools

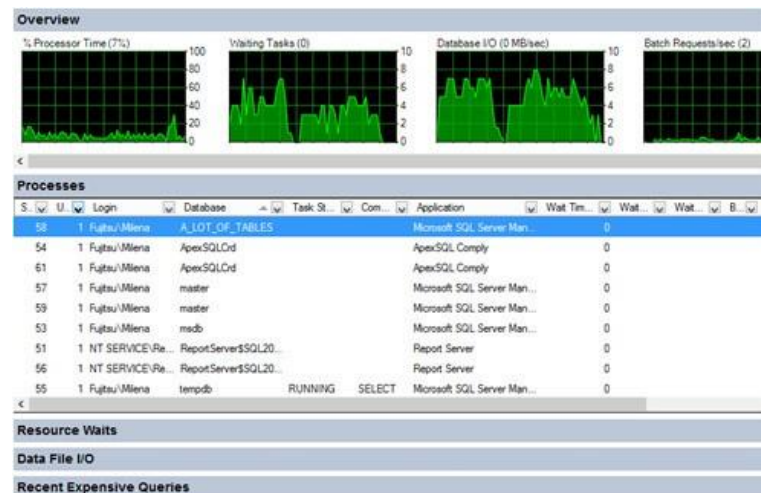
- Open Activity Monitor.
- SQL Trace.
- SQL Server Profiler.
- Database Engine Tuning Advisor DTA.
- DBCC (Transact-SQL).
- Dynamic Management Views DMVs.
- System Stored Procedures (Transact-SQL).
- SQL Server Extended Events.
- SQL Server Data Collector.
- Query Store
- Performance Dashboard
- Monitor Resource Usage (System Monitor).

2.1. Open Activity Monitor (SQL Server Management Studio)

- هي أداة تحوي واجهة رسومية GUI, تقوم بعرض عدة معلومات متعلقة بال SQL Server و ال DB. و عند ضبط زمن التحديث بأقل من 10 ثواني سيتم التأثير على أداء السيرفر.
- تحوي على خيار Active Expensive Queries الذي يتيح إستهلاك كافة الموارد
- كما يمكننا من فلترة النتائج على DB واحدة فقط من الموجودات في ال SQL Server
- يوفر خيار غير موجود لدى باقي الأدوات و هو إنهاء ال process أو ال Query التي تستهلك موارد كثيرة

Activity Monitor runs queries on the monitored instance to obtain information for the Activity Monitor display panes. When the refresh interval is set to less than 10 seconds, the time that is used to run these queries can affect server performance. It can kill processes, but a user must be a member of the sysadmin or processadmin fixed server roles.

Can open by CTRL + ALT + A (in 2017)



2.2. SQL Trace

- يقوم ب Capture لجميع ال Events التي تحدث على مستوى السيرفر
- لتفعيله نقوم بكتابته على شكل كود مكون من مجموعة الإستدعائات لل system stored procedures مع تحديد أرقام ال Events التي نريد أن يتم لها capturing (يجب أن نكون نعرف أرقام ال evets مسبقاً)

```

Ch06. Performance Monitoring and Tuning.sql - SQL (SQL\Administrator (109)) - not connected -... Quick Launch (Ctrl+Q)
File Edit View Query Project Tools Window Help
New Query
SVU
Execute
Stop the trace with sp_trace_setstatus.*/

---Create a trace by using sp_trace_create.
declare @trace_id int
declare @tracefile nvarchar(200)='D:\SVU\BAIT\IIS404\Demo\trace2.trc'
exec sp_trace_create @trace_id output,6,@tracefile
declare @trace_id int trace_getinfo(null) --to get tracId suplied by system
declare @tracefile nvarchar(200)='D:\SVU\BAIT\IIS404\Demo\trace2.trc'
exec sp_trace_create @trace_id output,6,@tracefile
select * from sys.fn_trace_getinfo(null) --to get tracId suplied by system
--Add events with sp_trace_setevent.
declare @trace_id int --46 is object created, 11 is userid
exec sp_trace_setevent 2 ,46,11,1
--Start the trace with sp_trace_setstatus.
exec sp_trace_setstatus 2,1
-- Now open the file
use SVU
create table tablena(id int)

133 %
Messages
A transport-level error has occurred when receiving results from the server. (provider: Shared Memory Provider)
Completion time: 2021-06-15T22:10:34.2038814+03:00
133 %

```

- بعد إستعماله صعب لذلك قامت ميكروسوفت بإستبداله فيما بعد بأداة أخرى ذات واجهة رسومية إسمها SQL Server Profiler و نصحت جميع المستخدمين بالتوقف عن إستعمال ال Trace

In SQL Trace, events are gathered if they are instances of event classes listed in the trace definition.

These events can be filtered out of the trace or queued for their destination. The destination can be a file or SQL Server Management Objects (SMO), which can use the trace information in applications

that manage SQL Server.

Microsoft SQL Server provides Transact-SQL system stored procedures to create traces on an instance of the SQL Server Database Engine. These system stored procedures can be used from within your own applications to create traces manually, instead of using SQL Server Profiler.

2.3. SQL Server Profiler

- يمكن بواسطته الفترة لتضمين Events معينة عن DB معينة
- نستطيع حفظ النتائج الخاصة به ك Trace File, و من ثم فتحها بواسطة الأداة Database Engine Tuning Advisor DTA التي تقوم بتحليل جميع النتائج ضمن هذا الملف

A graphical user interface to SQL Trace for monitoring an instance of the Database Engine or Analysis Services. You can capture and save data about each event to a file or table to analyze later. For example, you can monitor a production environment to see which stored procedures are affecting performance by executing too slowly. Open it from tools → SQL Server Profiler.

2.4. Database Engine Tuning Advisor DTA

- يقوم بتحليل ملفات ال Trace File أو أداء ال DB و إعطاء نصائح بناء عليها
- كما يمكننا تحديد أي Query لدينا بالضغط عليها باليمين و إختيار Analyze with Database Engine Tuning Advisor ليتم تحليلها بواسطته
- من النصائح التي يعطيها ممكن أن ينصح باستخدام Clustered Index لبعض الحقول مثلا مع إمكانية توفير الكود لإنشاء هذا ال Clustered Index في حال لا نعرف كيف
- لا ينصح باستعماله في البيئة الحقيقية للعمل, لأنه يقوم بإنشاء Indexes افتراضية (Hypothetical Indexes) لا يتم إزالتها إلا يدويا

Analyzes databases and makes recommendations that you can use to optimize query performance. Can be used to select and create an optimal set of indexes, indexed views, or table partitions without

having an expert understanding of the database structure or the internals of SQL Server. Using the DTA, you can perform the following tasks:

- Troubleshoot the performance of a specific problem query
- Tune a large set of queries across one or more databases
- Perform an exploratory what-if analysis of potential physical design changes
- Manage storage space

CAUTION

You should not run the Database Engine Tuning Advisor directly against a production server, because it can leave behind hypothetical indexes and statistics that can unknowingly persist. These will require additional resources to maintain. Use the `is_hypothetical` column in the `sys.indexes` system view to find hypothetical indexes for manual removal.

2.5. DBCC (Transact-SQL)

- هي **Database Console Command**, هي أداة تقوم بفحص الإتساقية الإفتراضية و الفعلية لل DB
- يتم تصنيف الإستعلامات الخاصة بهذه الأداة ضمن المجموعات التالية في الصورة:

Command category	Perform
Maintenance	Maintenance tasks on a database, index, or filegroup.
Miscellaneous	Miscellaneous tasks such as enabling trace flags or removing a DLL from memory.
Informational	Tasks that gather and display various types of information.
Validation	Validation operations on a database, table, index, catalog, filegroup, or allocation of database pages.

DBCC (Database Console Command) statements enable you to check performance statistics and the logical and physical consistency of a database. The Transact-SQL programming language provides DBCC statements that act as Database Console Commands for SQL Server.

Database Console Command statements are grouped into the following categories:

Command category	Perform
Maintenance	Maintenance tasks on a database, index, or filegroup.
Miscellaneous	Miscellaneous tasks such as enabling trace flags or removing a DLL from memory.
Informational	Tasks that gather and display various types of information.
Validation	Validation operations on a database, table, index, catalog, filegroup, or allocation of database pages.

- من أهم التعليمات الخاصة بال DBCC هي:

2.5.1. DBCC CHECKDB

- تقوم بالتحقق من الحالة الافتراضية و الفيزيائية لجميع الأغراض في ال DB (إستخدامها سابقا في عملية ضرب page من ال DB للتحقق من حالة جميع الصفحات)
عن طريق التحقق من مجموعة نقاط منها:

Checks the logical and physical integrity of all the objects in the specified database by performing the following operations:

- Runs DBCC CHECKALLOC on the database.
 - Checks the consistency of disk space allocation structures for a specified database.
- Runs DBCC CHECKTABLE on every table and view in the database.
 - Checks the integrity of all the pages and structures that make up the table or indexed view
- Runs DBCC CHECKCATALOG on the database.
 - Checks for catalog consistency within the specified database. The database must be online.
- Validates the contents of every indexed view in the database.
- Validates link-level consistency between table metadata and file system directories and files when storing varbinary (max) data in the file system using FILESTREAM.
- Validates the Service Broker data in the database.

2.5.2. DBCC - DBCC SQLPERF

- توفر معلومات عن حجم ال **log** في كل ال **Databases** مع إعطاء نسبة عن إستخدام القرص
- بالرغم من منفعتهما إلا أن ميكروسوفت تنصح بعدم إستخدامها و إستخدام **sys.dm_db_log_space_usage** حيث يقوم بتوفير نفس المعلومات و لكن خاصة بال **DB** الحالية فقط مع حجم ال **log** المستخدم من آخر **backup**

- Provides transaction log space usage statistics for all databases.
- Use sys.dm_db_log_space_usage DMV instead.

2.5.3. DBCC FREESYSTEMCACHE

- تقوم بتفضية جميع المساحة الغير مستخدمة في ال **cash** مما يتيح مساحة إضافية في ال **RAM**
- Releases all unused cache entries from all caches. The SQL Server Database Engine proactively cleans up unused cache entries in the background to make memory available for current entries. However, you can use this command to manually remove unused entries from all caches or from a specified Resource Governor Pool cache.

2.6. Dynamic Management Views (DMVs)

- تستخدم لمراقبة أداء السيرفر و حالته

Dynamic Management Views (DMVs) can be used to monitor SQL Server performance and troubleshoot issues.

- من تعليماته **sys.dm_exec_query_stats** التي تقوم بعرض جميع ال **Queries** التي يتم تنفيذها حالياً في الواجهة و في ال **background**

sys.dm_exec_query_stats – returns a row for every query statement in a cached query plan. It provides information about average, minimal, maximal, and total processor time used by the plan, along with other useful information for performance analysis

و تعليمة **sys.dm_exec_sessions** التي تقوم بعرض جميع ال **sessions** المفتوحة مع معلومات عنها

sys.dm_exec_sessions – returns a row for every session on the queried SQL Server instance, along with details such as the name of the program that initiated the session, session status, SQL Server login, various time counters, and more

و تعليمة **sys.dm_exec_requests** التي تقوم بعرض جميع ال **Requests** (عمليات القراءة و الكتابة) التي تتم حالياً مع الحالة الخاصة بها مثل **Running** أو **Suspended** أي بانتظار فك القفل عن البيانات المراد إستعمالها

sys.dm_exec_requests – returns a row for every user and system request being executed on the SQL Server instance. To find the blocked requests, search for the requests where the status column value is 'suspended'.

و تعليمة **sys.dm_db_log_stats** التي تقوم بعرض حالة ال **Log** الخاصة بال **DB** (عدد ال **VLF** و حجم كل منها و عدد النشطة منها)

sys.dm_db_log_stats - This DMV returns information about the transaction log files. The information includes the recovery model of the database.

```
select
    db_NAME(database_id) dbname,
    recovery_model,
    current_vlf_size_mb,
    total_vlf_count,
    active_vlf_count,
    active_log_size_mb,
    log_truncation_holdup_reason,
    log_since_last_checkpoint_mb
from
    sys.dm_db_log_stats(5)
```

	dbname	recovery_model	current_vlf_size_mb	total_vlf_count	active_vlf_count	active_log_size_mb	log_truncation_holdup_reason	log_since_last_checkpoint_mb
1	Python2017	FULL	1.9375	4	1	0.187988	NOTHING	0.187988

total log size

It is this easy to project the data for the entire database by cross-joining with the sys.databases system view. The derived output gives very useful information about the database and the log structure.

	name	recovery_model	current_vlf_size_mb	total_vlf_count	active_vlf_count	active_log_size_mb	log_truncation_holdup_reason	log_since_last_checkpoint_mb
1	master	SIMPLE	0.25	8	1	0.182617	NOTHING	0.182617
2	tempdb	SIMPLE	1.9375	4	1	0.296875	NOTHING	0.296875
3	model	FULL	1.875	6	1	0.060058	NOTHING	0.060058
4	msdb	SIMPLE	0.25	3	1	0.133789	NOTHING	0.133789
5	Python2017	FULL	1.9375	4	1	0.187988	NOTHING	0.187988

و تعلیمة **sys.dm_db_log_info** تعطينا معلومات عن كل VLF على حدى بعكس القبلها التي تقوم بعرض معلومات عن جميع ال VLFs بشكل كلى

sys.dm_db_log_info - The sys.dm_db_log_info also deals with the log files. It requires the database ID for input. This view specifically looks at virtual log files or VLFs. These make up the transaction log of the database and having a large number of VLFs can negatively impact the startup and recovery time of your database. With this view, we can see how many VLFs your database currently has, along with their size and status. This management view replaces the database console command, dbcc log info.

و تعلیمه sys.dm_db_tuning_recommendations

sys.dm_db_tuning_recommendations - The new DMV returns the detailed tuning recommendations information. This requires an enabling of the automatic tuning option. If the database engine identifies the plan that is causing regression then a suggestion will be made for fixing it using this dmv.

و تعلیمه sys.dm_os_host_info التي تقوم بعرض معلومات عن ال Host لمستضيف لل SQL Server

sys.dm_os_host_info

```
select * from sys.dm_os_host_info
```

host_platform	host_distribution	host_release	host_service_pack_level	host_sku	os_language_version
1	Linux	CentOS Linux	7	NULL	0

host_platform	host_distribution	host_release	host_service_pack_level	host_sku	os_language_version
1	Windows	Windows Server 2012 R2 Standard	6.3	7	1033

و تعلیمه sys.dm_db_file_space_usage التي تقوم بعرض معلومات عن التغييرات في الصفحات منذ آخر backup

sys.dm_db_file_space_usage - Returns space usage information of the database. The newly added column `modified_extent_page_count` gives us the flexibility to manage backups. The modified page count can be used to track differential changes in the database since the last full backup to decide if a differential backup is beneficial. The column is the total number of pages modified in the allocated extents of the file since the last full database backup

```
BACKUP DATABASE Python2017 to disk='f:\PowerSQL\Python2017.bak'

SELECT CAST(ROUND((SUM(modified_extent_page_count)*100.0)/SUM(allocated_extent_page_count),2)
as decimal(10,2)) as [Diff %]
from sys.dm_db_file_space_usage

update [dbo].[SQLShack]
set name='SQLShack_1'
The following query highlights how much data has been changed since the last full backup
SELECT CAST(ROUND((SUM(modified_extent_page_count)*100.0)/SUM(allocated_extent_page_count),2)
as decimal(10,2)) as [Percentage change since last backup]
from sys.dm_db_file_space_usage
```

Percentage Diff changes since last backup
1 14.55

2.7. System Stored Procedures

- يتم استخدامها من قبل ال DMVs و ال DBCCs

Many administrative and informational activities can be performed by using system stored procedures. Examples:

- Database Maintenance Plan Stored Procedures

sp_add_maintenance_plan, sp_delete_maintenance_plan_db, sp_add_maintenance_plan_db,
sp_delete_maintenance_plan_job, sp_add_maintenance_plan_job, sp_help_maintenance_plan,
sp_delete_maintenance_plan

- Database Engine Stored Procedures
- SQL Server Profiler Stored Procedures

2.8. SQL Server Extended Events

منذ إصدار الـ **SQL Server Extended Events** قامت ميكروسوفت بنصيحة المستخدمين باستخدامه عوضاً عن الـ **profilers**, بسبب إستهلاكه القليل للموارد (**Light Weight**) مع إمتلاكه تقريبا لنفس قدرة المراقبة مثل مراقبة الـ **most expensive queries, latches, deadlocks, locking**

حيث أن الإقفال المتبادل (**Dead Lock**) يحدث عندما تنتظر مناقلتان كل منهما الأخرى لكي تحرر قفلها عن عنصر معطيات معين ولحل هذه المشكلة يقوم الـ **SQL Server** بإنهاء إحدى هذه المناقلات بشكل قسري

SQL Server Extended Events has a highly scalable and highly configurable architecture that allows users to collect as much or as little information as is necessary to troubleshoot or identify a performance problem.

2.8.1. Benefits of SQL Server Extended Events

Extended Events is a light weight performance monitoring system that uses very few performance resources. Extended Events provides two graphical user interfaces (New Session Wizard and New Session) to create, modify, display, and analyze your session data.

It allows collecting information useful for troubleshooting SQL Server performance issues, and enables finding the most expensive queries, latch, deadlock, and blocking causes, troubleshooting excessive processor usage. To use Extended Events for performance monitoring, determine which events you want to monitor, and create a session using SQL Server Management Studio options, or T-SQL.

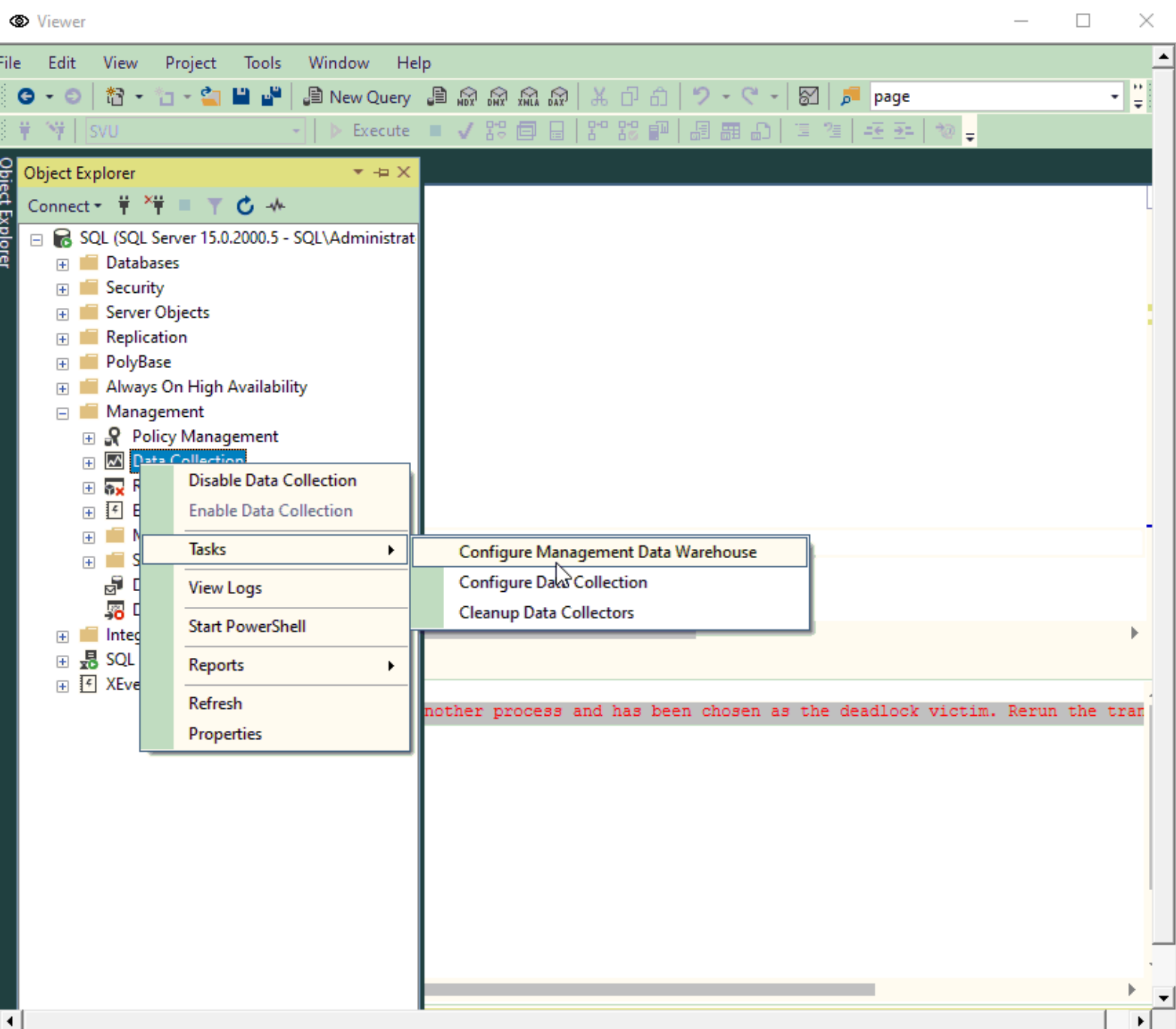
2.8.2. Extended Events Concepts

SQL Server Extended Events (Extended Events) builds on existing concepts, such as an event or an event consumer, uses concepts from Event Tracing for Windows, and introduces new concepts.

2.9. SQL Server Data Collector

يقوم بإنشاء Management Data Warehouse عن معلومات مراقبة الأداء الخاصة بهذه ال DB و DBs أخرى

حيث لاستخدامها نقوم ب **Enable** لل **Data Collection**
ثم من ال **Tasks** الخاص بها نختار **Configure Management Data Warehouse**



و بعد الإنتهاء من إنشائه, نقوم من نفس القائمة بإختيار **Configure Data Collection** لإستخدامها في عرض المعلومات عن

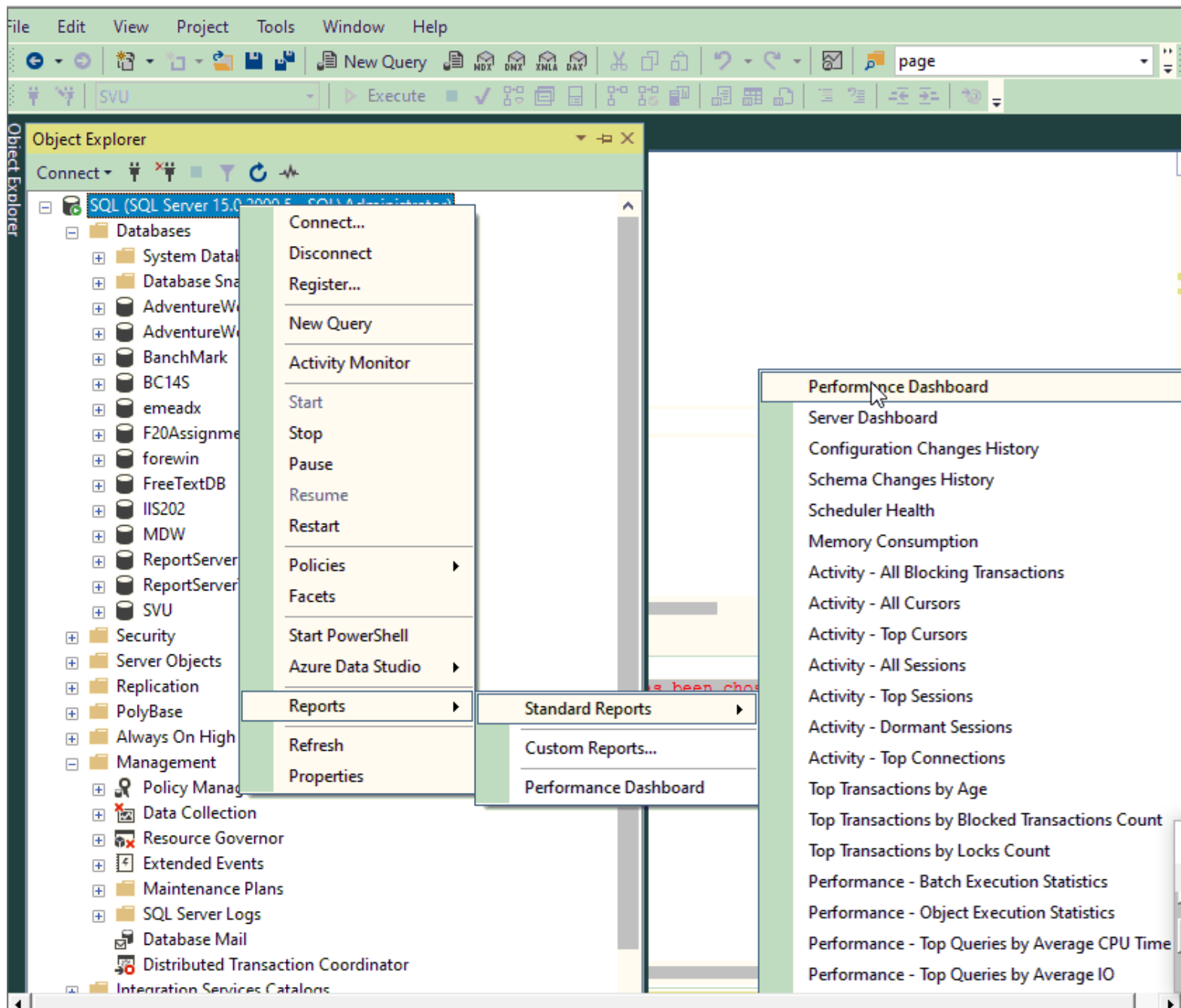
جميع التقارير الخاصة بالأداء لل DBs المختارة ضمنه

SQL Server Data Collector is another SQL Server Management Studio feature that can be used for SQL Server performance monitoring and troubleshooting.

Data Collector can collect performance metrics on multiple SQL Server instances and store them in a single repository. Data collection is possible only if SQL Server Agent is running and Management Data Warehouse is configured.

2.10. Query Store

- لإستخدامها:



تقوم بتوفير معلومات عن ال Execution Plane الخاص بال Query (مسار تنفيذها)

مع إخبارنا عند تنفيذ ال Query بمسار مختلف مع إمكانية المقارنة بينهما و إمكانية إجبار ال Query لإستخدام أحد المسارات

بشكل دائم لتحقيق أداء أعلى (و هذه الميزة غير موجودة غير في هذه الأداة)

Query Store

- The SQL Server Query Store feature provides you with insight on query plan choice and performance. It simplifies performance troubleshooting by helping you quickly find performance differences caused by query plan changes.
- Query Store automatically captures a history of queries, plans, and runtime statistics, and retains these for your review. It separates data by time windows so you can see database usage patterns and understand when query plan changes happened on the server.

كما تقوم بتوفير Reports عن: Regressed Queries – الإستعلامات المترجع أدائها

Overall Resource Consumption – إستهلاك الموارد الكلي من قبل ال Execution Plane لكل Query

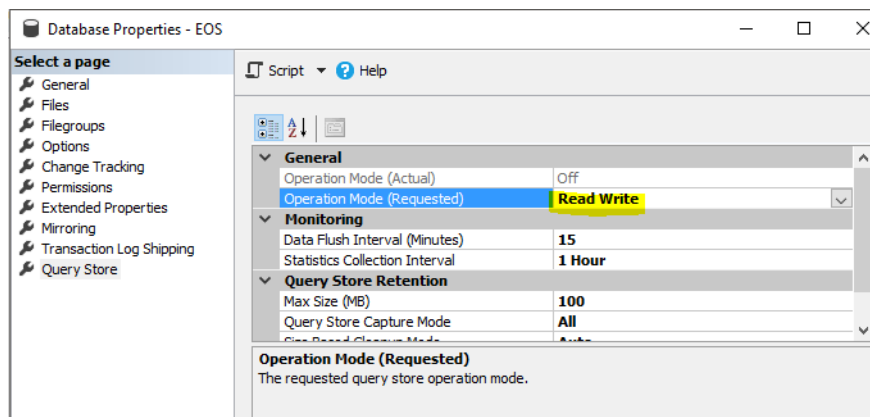
The SQL Server Query Store feature provides you with insight on query plan choice and performance. It simplifies performance troubleshooting by helping you quickly find performance differences caused by query plan changes. Query Store automatically captures a history of queries, plans, and runtime statistics, and retains these for your review. It separates data by time windows so you can see database usage patterns and understand when query plan changes happened on the server.

2.10.1. Enabling the Query Store

Query Store is not active for new databases by default.

1. Use the Query Store Page in SQL Server Management Studio

- In the **Database Properties** dialog box, select the **Query Store** page.
- In the Operation Mode (Requested) box, select Read Write.



2. By Using T-SQL

SQL

```
ALTER DATABASE AdventureWorks2012 SET QUERY_STORE (OPERATION_MODE = READ_WRITE);
```

2.10.2. Information in the Query Store

Execution plans for any specific query in SQL Server typically evolve over time due to a number of different reasons such as statistics changes, schema changes, creation/deletion of indexes, etc. The procedure cache (where cached query plans are stored) only stores the latest execution plan. Plans also get evicted from the plan cache due to memory pressure. As a result, query performance regressions caused by execution plan changes can be non-trivial and time consuming to resolve.

Common scenarios for using the Query Store feature are:

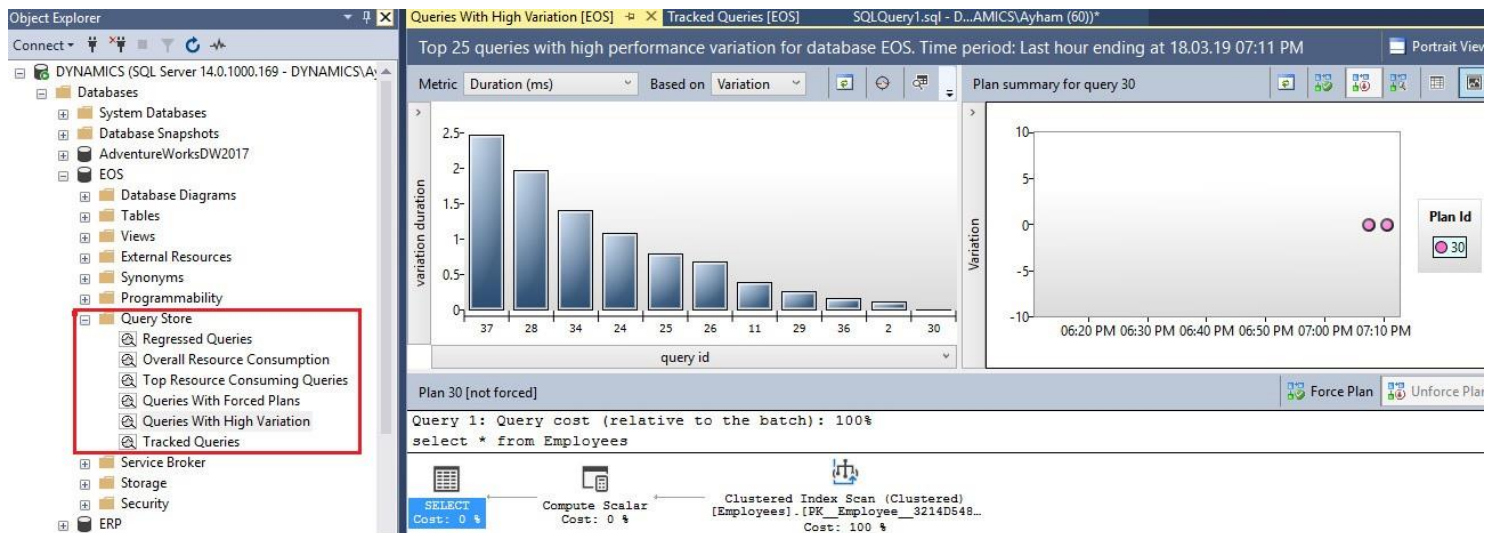
- Quickly find and fix a plan performance regression by forcing the previous query plan. Fix queries that have recently regressed in performance due to execution plan changes.
- Determine the number of times a query was executed in a given time window, assisting a DBA in troubleshooting performance resource problems.
- Identify top n queries (by execution time, memory consumption, etc.) in the past x hours.
- Audit the history of query plans for a given query.
- Analyze the resource (CPU, I/O, and Memory) usage patterns for a particular database.
- Identify top n queries that are waiting on resources.
- Understand wait nature for a particular query or plan.

The query store contains three stores:

- A plan store for persisting the execution plan information.
- A runtime stats store for persisting the execution statistics information.
- A wait stats store for persisting wait statistics information.
- The following query returns information about queries and plans in the query store.

```
SELECT Txt.query_text_id, Txt.query_sql_text, Pl.plan_id, Qry.*
FROM sys.query_store_plan AS Pl
INNER JOIN sys.query_store_query AS Qry
    ON Pl.query_id = Qry.query_id
INNER JOIN sys.query_store_query_text AS Txt
    ON Qry.query_text_id = Txt.query_text_id ;
```

After enabling Query Store, a set of performance monitoring could be obtained as the following:



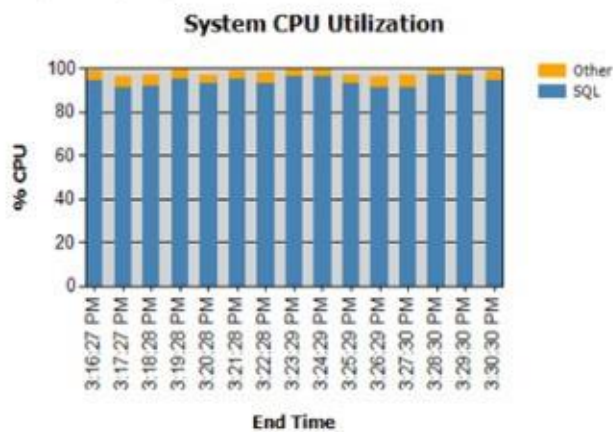
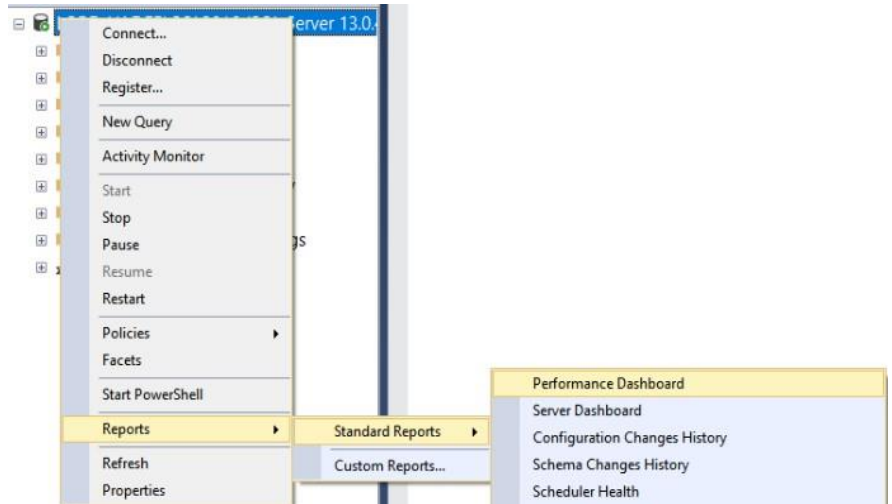
2.11. Performance Dashboard

SQL Server Management Studio version 17.2 and later includes the Performance Dashboard. This dashboard was designed to visually provide fast insight into the performance state of SQL Server (Starting with SQL Server 2008).

The Performance Dashboard helps to quickly identify whether SQL Server is experiencing a performance bottleneck. And if a bottleneck is found, easily capture additional diagnostic data that may be necessary to resolve the problem. Some common performance problems which the Performance Dashboard can help identify include:

- CPU bottlenecks (and what queries are consuming the most CPU)
- I/O bottlenecks (and what queries are performing the most IO)
- Index recommendations generated by the Query Optimizer (missing indexes)
- Blocking
- Resource contention (including latch contention)

The Performance Dashboard also helps to identify expensive queries that may have been executed before, and several metrics are available to define high cost: CPU, Logical Writes, Logical Reads, Duration, Physical Reads, and CLR Time.



Current Activity

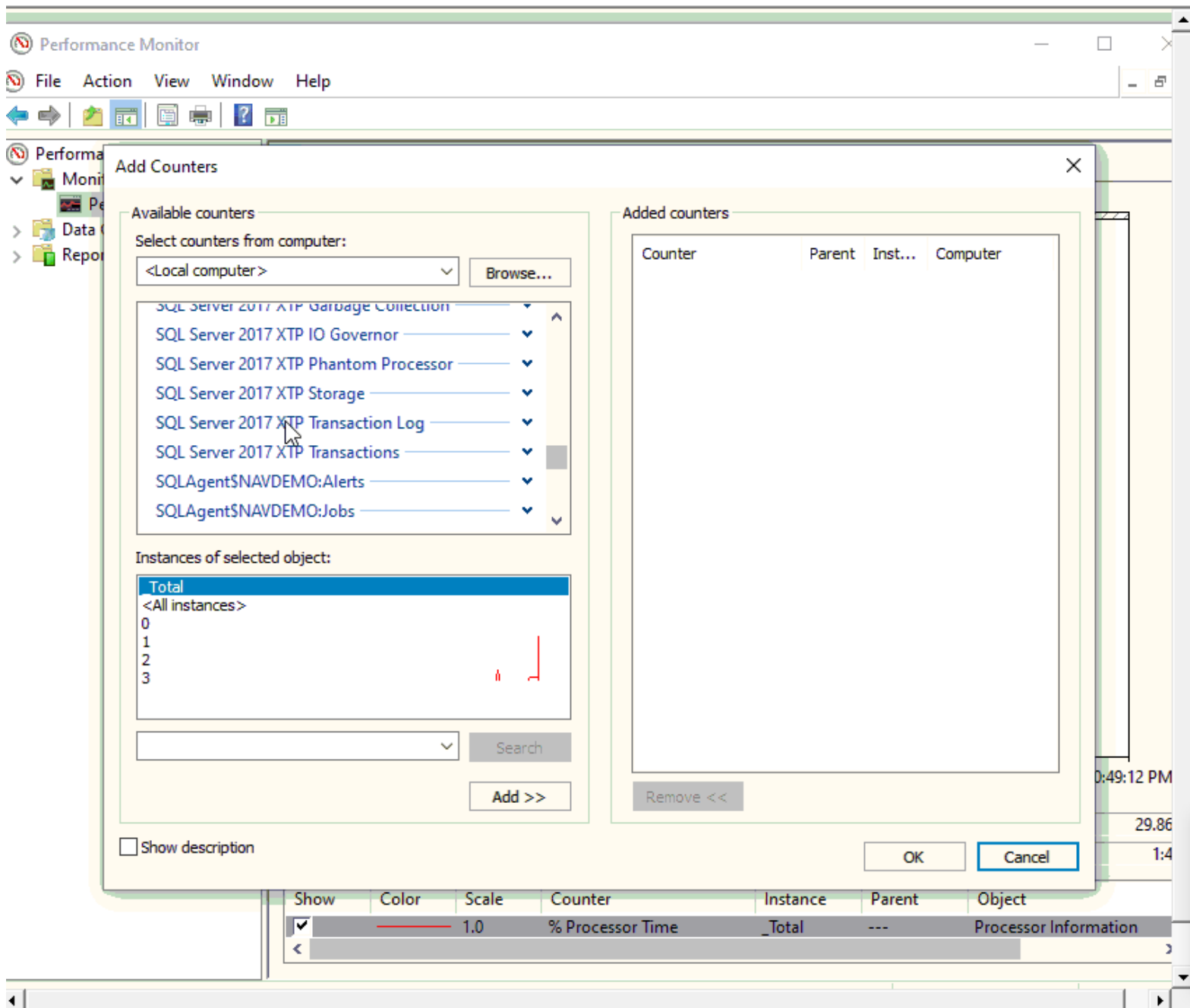
	User Requests	User Sessions
Count	27	32
Elapsed Time (ms)	4573004	741818
CPU Time (ms)	2043203(44.68%)	101108(13.63%)
Wait Time (ms)	2529801(55.32%)	640710(86.37%)
Cache Hit Ratio	100.000%	98.313%

Historical Information

Waits	IO Statistics
Latches	
Expensive Queries	
By CPU	By Duration
By Logical Reads	By Physical Reads
By Logical Writes	By CLR Time
Miscellaneous Information	
Active Traces	1
Active Xevent Sessions	4
Databases	16
Missing Indexes	11

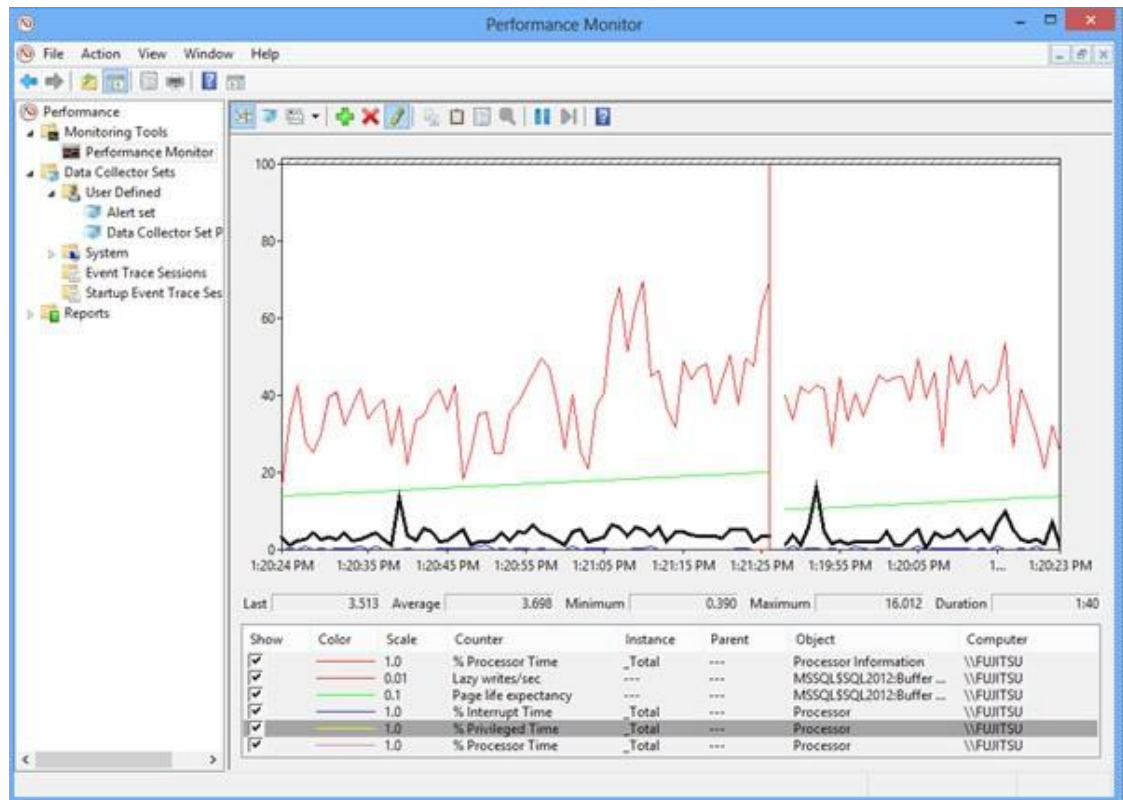
2.12. Windows Performance Monitor

هو أداة موجودة في ويندوز، مشكلته أنه يستهلك كثير من الموارد ليحلل إستهلاك الموارد و أداء ال DB حيث لتفعيل مراقبة ال SQL Server نقوم بإضافة Counter و تحديد نوعها ب SQL Server :



و تستخدم من قبل ال System Administrator أكثر من ال Database Administrator -

Windows Performance Monitor shows useful information in comprehensive real-time graphs and can save historical data for a long time, so it can be used for later analysis. Its disadvantages are that it cannot provide sufficient information for deep analysis nor show a trend line or a threshold in the graph.



References

LeBlanc. P, Assaf. W, Jackson. B, Curnutt M; “**SQL Server 2017 Administration Inside Out**”, Microsoft Press (2018).

Microsoft SQL Server Mission-Critical Performance, **Technical White Paper**, Microsoft (2016)

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